

May 2025

011602

B.Tech. (Sixth Semester)
Data Analysis Using Python (PCC-IT-601-A-2024)

Time : 3 Hours]

[Maximum Marks : 75

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) Write a python code to concatenate two lists. 1.5
- (b) Differentiate between numpy arrays and lists. 1.5
- (c) What is `__init__` in Python ? 1.5
- (d) Find the variance for the following set of data representing trees heights in feet : 1.5
3, 21, 98, 203, 17, 9.
- (e) Explain overfitting and underfitting of data in data analytics. 1.5

P.T.O.

- (f) What is Data visualization ? 1.5
- (g) How to access first few rows of a Dataframe in Pandas ? 1.5
- (h) Differentiate between inferential and Descriptive Statistics. 1.5
- (i) List 3 magic methods. 1.5
- (j) What is an abstract class ? 1.5

Part B

2. (a) What is hypothesis testing ? Explain the various steps used in hypothesis testing. 10
- (b) What is numpy used for ? WAP to find mean, median and standard deviation of a numpy array using Python. 5
3. (a) What is a file ? Describe the various modes of operation in a file. 5
- (b) A researcher wants to know if the average test scores of students from three different teaching methods are the same or not. The test scores from three groups of students who were taught using different methods are shown below :

Group 1(Traditional method) : 56. 58. 60.
62. 64

Group 2(Online method) : 68. 70. 72. 74. 76

Group 3 (Blended method) : 75. 78. 80.
82. 85

Make an analysis of the variance of the data
using ANOVA. 10

4. (a) Explain the following using syntax in context
of numpy : 10

(i) Numpy array attributes

(ii) Splitting arrays

(iii) Dtype objects

(iv) Boolean indexing.

(b) Explain in detail the logistic regression
approach. 5

5. (a) How does classification differ from
clustering? Explain. 5

(b) Write a program in Python to demonstrate
Exception Handling 10

6. (a) What is a Dataframe in Python ? Create a dictionary that stores the mobile names list as value for 'Mobiles' key and their price list as value for 'Price' key. Create DataFrame from this dictionary. 10

(b) What are the various data visualization techniques ? Explain. 5

7. Write short notes on the following : 15

(a) Packages

(b) Probability distribution types

(c) Inheritance types in Python.

